Ratios, Proportions & Percentages

Section 1: Direct & Inverse Proportions

1 If 6 workers can paint a fence in 10 hours, how many hours would it take 15 workers working at the same rate?

a. 4 hours

b. 5 hours

c. 6 hours

d. 8 hours

2 A car travels 240 km using 20 liters of fuel. How many liters are required to travel 360 km at the same rate?

a. 30 liters

b. 28 liters

c. 32 liters

d. 35 liters

3 If $a : b = 2 : 3$ and $b : c = 4 : 5$, what is the ratio $a : c$?

a. $8 : 15$

b. $2 : 5$

c. $6 : 15$

d. $8 : 12$

Section 2: Percentages, Discounts & Markups

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4 A jacket originally priced at 150 SAR is discounted by 20%. What is the sale price?

a. 120 SAR

b. 110 SAR

c. 125 SAR

d. 130 SAR

5 The price of an item was increased by 20% and then decreased by 20%. What is the net percentage change in price?

a. 4% decrease

b. 0%

c. 2% decrease

d. 4% increase

6 If 25% of a number x is 45, what is 60% of x?

a. 108

b. 90

c. 120

d. 135

Section 3: Speed, Distance & Average Velocity

7 A car travels from City A to City B at 60 km/h and returns along the same route at 40 km/h. What is the average speed for the round trip?

a. 48 km/h

b. 50 km/h

c. 45 km/h

d. 52 km/h

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8 Two trains leave the same station in opposite directions at speeds of 70 km/h and 80 km/h. In how many hours will they be 450 km apart?

a. 3 hours

b. 2.5 hours

c. 3.5 hours

d. 4 hours

9 A cyclist travels 15 km in 45 minutes. What is their speed in kilometers per hour?

a. 20 km/h

b. 18 km/h

c. 25 km/h

d. 15 km/h

Section 4: Work Rates & Shared Labor

10 Pipe A can fill a tank in 6 hours, while Pipe B can fill it in 3 hours. If both pipes are opened together, how long will it take to fill the tank?

a. 2 hours

b. 2.5 hours

c. 1.5 hours

d. 3 hours

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11 Ahmed can complete a job in 4 days, and Khalid can complete it in 6 days. If they work together, how many days will the job take?

a. 2.4 days

b. 2.5 days

c. 3 days

d. 2 days

12 A tank has a leak that can empty it in 12 hours. If an inlet pipe fills it in 4 hours, how long will it take to fill the tank with the leak open?

a. 6 hours

b. 5 hours

c. 8 hours

d. 4.5 hours

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